

PAPER_229: Pillars of Creation (Eagle Nebula M16) – MUGE with Decaying Photoevaporation Erosion Factor $E(t)$

Author: Daniel T. Murphy **Framework:** UQFF v4.8 (Star-Magic) **Session:** 58 (grok_share_8d951e12.txt extraction – Doc 7) **Date:** March 2026 **Classification:** Novel MUGE Term – Decaying Erosion Suppression ($1 - E(t)$) on Base Gravity **Status:** Proof-Quality Whitepaper

Abstract

The Pillars of Creation in the Eagle Nebula (M16, NGC 6611, $\sim 6,500$ ly) are modelled with a 9-term MUGE incorporating a uniquely novel term: a decaying photoevaporation erosion factor $E(t) = E_0 e^{-t/\tau_e}$ applied to the base gravity as a multiplicative suppression ($1 - E(t)$). This is physically distinct from the Bubble Nebula (PAPER_221) which uses ($1 + E(t)$) for shell compression enhancement. The sign physically distinguishes systems where EUV radiation removes mass (erosion – less gravity) versus compresses it (shock + more gravity).

1. Physical System

The Pillars are evaporating proto-stellar columns irradiated by the NGC 6611 O-star cluster:

Parameter	Value
Distance	$\sim 6,500$ ly
Pillar height	$\sim 4 - 5$ ly
M_{init}	$100 M_{\odot}$ (pillar stellar cores)
M_{gas}	$10,000 M_{\odot}$ (column gas mass)
r	5 ly
B	$1 \mu\text{T}$
EUV source	NGC 6611 O-stars ($T_{eff} > 40,000$ K)
E_0	0.1 (10% peak erosion suppression)
τ_e	1 Myr

2. Decaying Erosion Factor (Novel)

2.1 Definition

$$E(t) = E_0 e^{-t/\tau_e}$$

2.2 Application to Base Gravity

$$a_{base} = \frac{GM(t)}{r^2} (1 + H_0 t) \left(1 - \frac{B}{B_{crit}} \right) (1 - E(t))$$

At $t = 0$: $E = E_0 = 0.1$ – base gravity suppressed by 10% (maximum erosion). At $t \gg \tau_e$: $E \rightarrow 0$ – gravity recovers fully as the pillar material disperses or the EUV source weakens.

2.3 Physical Interpretation

The factor $(1 - E(t))$ represents the fractional mass remaining in the gravitational region after photoevaporative mass loss. As EUV photons ablate the pillar surface, the effective gravitating mass decreases, reducing a_{base} .

3. Sign Comparison: Erosion vs Compression

System	Erosion/Expansion Term	Sign	Physical Process
Pillars of Creation (PA-PER_229)	$(1 - E(t)), E_0 = 0.1$	$\hat{a}''$	EUV ablation removes mass $\hat{a}''$ less gravity
Bubble Nebula (PA-PER_221)	$(1 + E(t)),$ growing	$+$	Shock compression $\hat{a}''$ more gravity
Orion Nebula (PA-PER_xxx)	None	N/A	Young, pre-dispersal

The sign of the erosion factor is physically determined by whether the process removes or adds mass to the gravitating region.

4. Canonical Result

At $t = 0.1$ Myr with $M(0) = 100.1M_{\odot}$:

$$a_{base} \approx \frac{G \times 1.989 \times 10^{32}}{(4.73 \times 10^{16})^2} \times (1 - 0.1 \times e^{-0.1}) \approx 5.93 \times 10^{-24} \times 0.905 \approx 5.36 \times 10^{-24} \text{ m/s}^2$$

5. Simulation Parameters

Parameter	Value
$t_{canonical}$	0.1 Myr
dt	0.001 Myr
M_{dot_factor}	$M_{gas}/M_{init} = 100$
τ_{SF}	5 Myr
E_0	0.1
τ_e	1 Myr

6. Calculator Class

```
class PillarsOfCreationErosionMUGECalculator(_CP3Calculator):
    """PAPER_229: Pillars of Creation â€” 9-term MUGE, decaying erosion (1-E(t)) on bas
    # Session 58 â€” grok_share_8d951e12.txt Doc 7
```

7. Conclusion

The decaying erosion factor $(1 - E_0 e^{-t/\tau_e})$ is a genuinely novel MUGE term. Combined with the sign-comparison against the Bubble Nebula's $(1 + E(t))$, this establishes a physically rigorous erosion/compression taxonomy within the UQFF nebular MUGE family: negative sign for ablative processes, positive for compressive ones.

Source: grok_share_8d951e12.txt Doc 7 (Pillars of Creation Erosion MUGE)

UQFF computed: MUGE buoyancy ratio $U_{bi}/F_U = [SSq] \times r^2/GM = 5.7e-1 \times 5.0e-4 = 2.85e-4$; compressed MUGE baseline $g = 5.4e-7 \text{ m/s}^2$ at r_{ISCO} .